

SituationReport

testdata_generator

User Manual

Generate synthetic Jira issue data for SituationReport

situation-report -- github.com/Jaegerfeld/situation-report

For Agile Coaches and PI Managers -- Version 0.99

1 What is the Testdata Generator?

The **Testdata Generator** creates synthetic Jira issue data in Jira REST API format. The generated files can be processed directly by **transform_data** — no real Jira instance required.

The generator simulates realistic workflow histories: issues traverse the defined stages, spend varying amounts of time in each stage, occasionally flow backwards, and close at a configurable rate.

Typical use cases:

- **Test a new installation:** Verify that `transform_data` and `build_reports` work correctly
- **Demos and presentations:** Realistic data without privacy concerns
- **Training:** Controlled learning environment with known data
- **Development:** Reproducible test data via the seed parameter

The Testdata Generator complements the **Get Data** module (still in development), which loads real data directly from Jira. For the manual export of real Jira data, see the **Get Data User Manual**.

2 Prerequisites and Start

2.1 Portable package (recommended)

The portable package includes Python already bundled — no separate installation required.

Operating system	Start file
Windows	TestdataGenerator.bat (double-click)
macOS	TestdataGenerator.command (right-click → Open)
Linux	./TestdataGenerator.sh

Note for Windows: If double-clicking shows an error about Python not being found, the ZIP file may have been blocked by Windows. Fix: right-click the ZIP → Properties → Security → Unblock → OK → extract again.

2.2 From source code (Python required)

```
python -m testdata_generator
```

2.3 The user interface

After launch, the main window shows input fields for all parameters. Only the workflow file is required — all other fields have sensible defaults.

testdata_generator 0.9.2

Optionen / Options

Workflow-Datei

Ausgabedatei (JSON)

Projekt-Key

Anzahl Issues

Von (YYYY-MM-DD)

Bis (YYYY-MM-DD)

Issue-Typen (Typ:Gewicht ...)

Completion-Rate (0-1)

To-Do-Rate (0-1)

Backflow-Prob. (0-1)

Seed (leer = zufällig)

Log

Fig. 1: Testdata Generator user interface

3 The Workflow File

The workflow file defines the stages of your Jira board and is the only required parameter. It uses the same format as `transform_data` — one file works for both modules.

3.1 Format

Each line describes a stage. Aliases (alternative status names in the Jira export) are separated by colons. Two special lines mark the start and end of active work:

```
CanonicalName:Alias1:Alias2
<First>FirstActiveStage
<Closed>ClosedStage
```

Line	Meaning
CanonicalName:Alias1:Alias2	Stage with one or more alternative Jira status names
StageName (no :)	Stage without aliases
<First>StageName	First active stage — where processing begins (cycle time starts here)
<Closed>StageName	Closed stage — marks an issue as completed

3.2 ART_A example workflow

The bundled example workflow `ART_A.txt` models a typical SAFe ART workflow:

```
Funnel:New:Open:To Do
Analysis:In Analysis:Estimated
Program Backlog:Ready for Dev:Backlog
Implementation:In Implementation:In Review:In Progress
Blocker
Validating on Staging:QA
Deploying to Production:Ready for Development
Releasing:Completed
Done:Canceled
<First>Analysis
<Closed>Releasing
```

Notes: 'Funnel' has four aliases — all four Jira status names are normalised to 'Funnel'. Analysis is the first active stage (cycle time begins here). Releasing is the closed stage. Done (after Releasing) also captures Canceled issues.

4 Parameter Reference

All parameters are available both in the GUI and on the command line.

Parameter	Default	Description
--workflow FILE	(required)	Workflow definition file (.txt)
--output FILE.json	<project>_generated.json	Output file path
--project KEY	TEST	Jira project key for generated issues
--issues N	100	Number of issues to generate
--from-date YYYY-MM-DD	2025-01-01	Earliest creation date
--to-date YYYY-MM-DD	2025-12-31	Latest transition date
--issue-types TYPE:W ...	Feature:0.6 Bug:0.3 Enhancement:0.1	Issue types with weights (sum need not equal 1)
--completion-rate FLOAT	0.7	Fraction of issues reaching the closed stage (0–1)
--todo-rate FLOAT	0.15	Fraction of open issues staying in early stages (0–1)
--backflow-prob FLOAT	0.1	Probability of a backward transition at each step (0–1)
--seed INT	(random)	Seed for reproducible output (optional)
--mean-cycle-days FLOAT	(none)	Target mean cycle time in days (lognormal); enables CT control
--std-cycle-days FLOAT	(30 % of mean)	Standard deviation of cycle time in days
--pattern PATTERN	none	Flow anti-pattern: none / triangle / flat_triangle / cluster / batch
--pi-duration-weeks INT	12	PI cycle length in weeks (for cluster/batch)

Tip: Reproducible results

With a fixed seed (e.g. `--seed 42`), the generator always produces the same JSON file for the same parameters — ideal for demos where all participants should see identical results.

4b Flow Anti-Pattern Modes

Since version 0.99, the Testdata Generator can simulate typical flow anti-patterns from Vacanti/Singh. Combine **--mean-cycle-days** with **--pattern**.

Pattern	Scatter plot appearance	Explanation
none	Even point cloud	Random cycle time without trend (default)
triangle	Triangle with right angle at bottom left	Cycle time increases linearly over time: $\text{mult} = 1 + 3 \times t$
flat_triangle	Triangle, but the rise flattens at top	Cycle time increase is damped by $\tanh(2.5 \times t)$
cluster	Vertical clusters of points at PI edges	CT lognormal (stable), but deliveries in last 2 weeks of each PI
batch	Vertical columns with variable CT	CT uniform 0.1× to 3× mean + PI clustering

Examples

```
# Triangle pattern
python -m testdata_generator \
  --workflow workflow.txt --issues 300 \
  --pattern triangle --mean-cycle-days 20 --std-cycle-days 6 \
  --output triangle.json
```

```
# Cluster of Dots (12-week PI)
python -m testdata_generator \
  --workflow workflow.txt --issues 300 \
  --pattern cluster --mean-cycle-days 30 \
  --pi-duration-weeks 12 --output cluster.json
```

```
# Batch Transfers
python -m testdata_generator \
  --workflow workflow.txt --issues 300 \
  --pattern batch --mean-cycle-days 30 \
  --pi-duration-weeks 12 --output batch.json
```

Note: **--pattern** requires **--mean-cycle-days**. Without **--mean-cycle-days** the pattern is ignored and the original min/max dwell-hour behaviour is used.

5 ART_A – Complete Example

This example shows the full pipeline from generator to finished report using the fictional ART_A project.

Step 1: Generate test data

```
python -m testdata_generator \  
    --workflow testdata_generator/workflow_ART_A.txt \  
    --project ART_A \  
    --issues 200 \  
    --from-date 2025-01-01 \  
    --to-date 2025-12-31 \  
    --issue-types Feature:0.6 Bug:0.2 Enabler:0.2 \  
    --completion-rate 0.75 \  
    --backflow-prob 0.08 \  
    --seed 42 \  
    --output ART_A_generated.json
```

Result: ART_A_generated.json with 200 issues — approximately 150 closed and 50 open across various stages. Seed 42 makes the output reproducible at any time.

Step 2: Process with transform_data

```
python -m transform_data ART_A_generated.json \  
    --workflow testdata_generator/workflow_ART_A.txt
```

Produces three Excel files: ART_A_generated_IssueTimes.xlsx, ART_A_generated_CFD.xlsx, and ART_A_generated_Transitions.xlsx.

Step 3: Create report with build_reports

```
python -m build_reports
```

Load the three Excel files from Step 2 in build_reports — the module generates flow diagrams, cycle time analyses, and metrics.

Expected output (seed 42, 200 issues):

- Issue keys: ART_A-0001 to ART_A-0200
- Issue types: ~120 Feature, ~40 Bug, ~40 Enabler
- ~150 issues with status 'Releasing' or 'Done'
- Creation dates distributed across 2025
- Occasional backward transitions (backflow-prob 0.08 = ~8 %)

6 Frequently Asked Questions (FAQ)

Difference between test data and real data?

For demos, training, and installation tests, test data is the better choice — no privacy concerns, reproducible, and fully controlled. For real flow analyses and decisions, actual data reflects the true workflow state.

For the manual export of real Jira data (API token, curl, pagination): See the Get Data User Manual.

7 Glossary

Term	Explanation
ART	Agile Release Train — a team-of-teams structure in SAFe
API	Application Programming Interface — programming interface
API token	Security key for accessing the Jira REST API; replaces the password in curl calls
Changelog	Jira log of all status changes for an issue (requires expand=changelog)
CFD	Cumulative Flow Diagram — shows the cumulative issue count per stage over time
Closed stage	The stage that marks an issue as completed (workflow.txt: <First>Closed<First>)
Cycle time	Time from the first active stage (<First>First<First>) to the closed stage
First stage	First active stage where cycle time measurement begins (workflow.txt: <First>First<First>)
Helper	SituationReport module for merging multiple Jira JSON files
Issue	A work item in Jira (Feature, Bug, Enabler, Story, ...)
JQL	Jira Query Language — query language for Jira searches, e.g. project=ART_A
JSON	JavaScript Object Notation — format of Jira API exports
Seed	Starting value for the random generator — same seed always produces the same output
Stage	One step in the workflow (corresponds to a Jira column or status name)
Workflow	Ordered sequence of stages that an issue passes through